

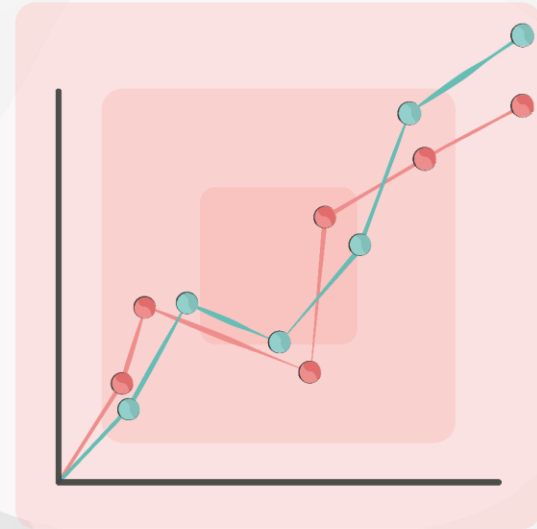


Data engineering

Module: Data pipelines

Topic: Data integration techniques

**Welcome to today's
webinar.**



Ice breaker

Discussion...

- How are you feeling today? Motivated, happy etc?
- What is your key takeaway from the e-learning topic?
- What is one key skill or insight you hope to gain from today's session?

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**Submit your responses to the
chat or turn on your
microphone**

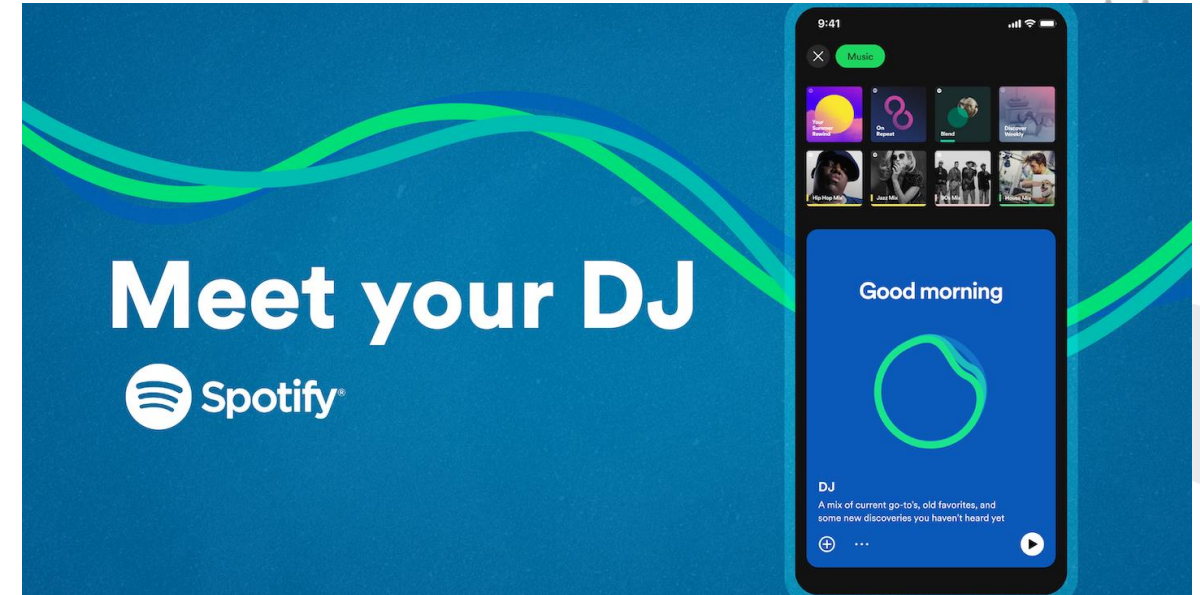
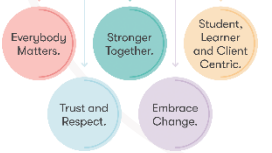


Spotify's Scalable Data Platform

Streaming, Cloud-Native, and Insight-Driven Integration...

- Migrated 1,200+ services and 20,000+ daily jobs to Google Cloud
- Uses Pub/Sub, Dataflow, and BigQuery for real-time analytics
- Supports billions of daily plays with low-latency data pipelines
- Enables personalised recommendations and artist insights
- Focuses on developer velocity and scalable, hybrid architecture

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The Spotify platform, Image source: [Spotify](#)

e-learning recap

Reflecting on your learning...

The e-learning for this topic, covered the following areas:

- **ETL** – Too slow for modern needs
- **Methods** – Flexible access via ELT, federation, etc.
- **Streaming** – Real-time insights
- **Architecture** – Simple, centralised, or scalable
- **Tools** – NiFi, Kafka, dbt, Talend, Airflow
- **Governance** – Validate and secure data
- **Hybrid** – Mix methods for resilience



- Do you have any questions about any of these areas?
- Did everything in the e-learning make sense?

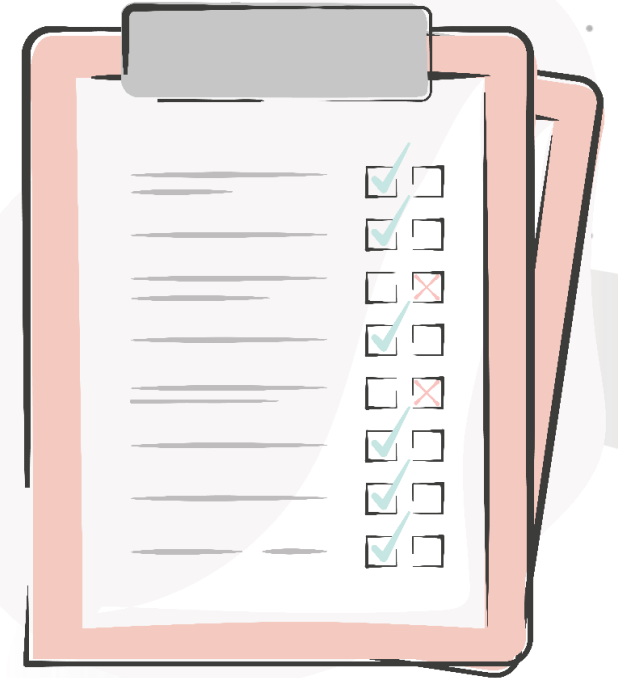
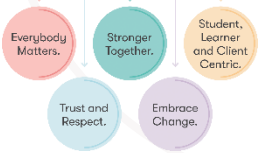


Webinar Agenda

Today, we will cover the following:

1. Welcome and Intro
2. Recap of core concepts
3. Practical lab
4. Summary
5. Q&A

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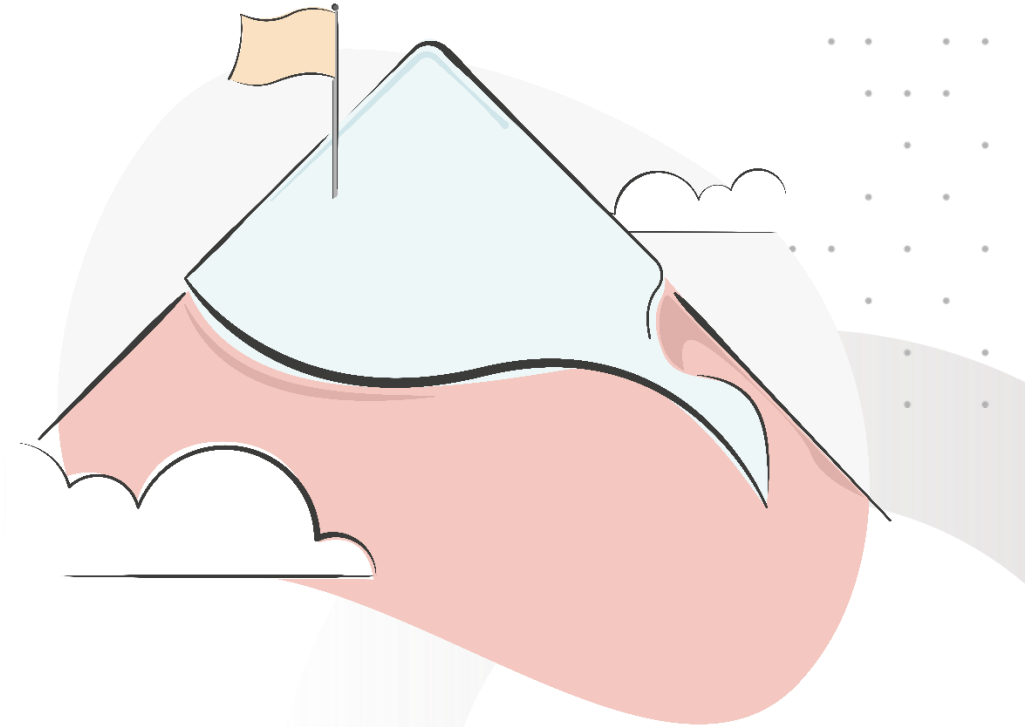
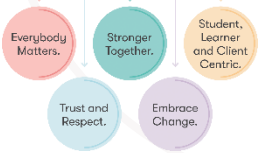


Session aim and objectives

By the end of this session, you should be able to:

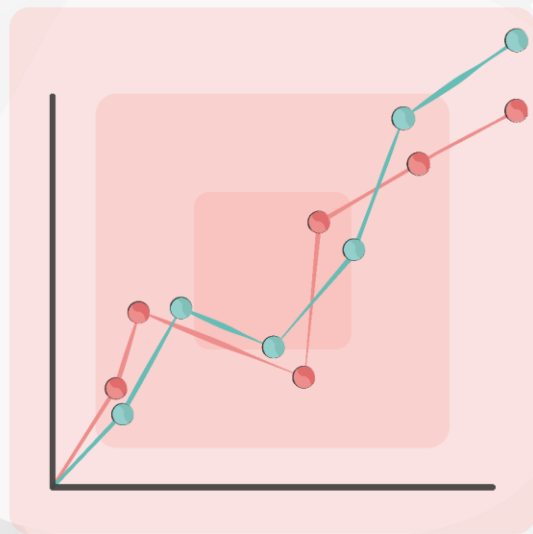
- Understand advanced integration methods beyond traditional ETL.
- Explore real - time and federated data integration approaches.
- Evaluate modern tools used in advanced integration scenarios.

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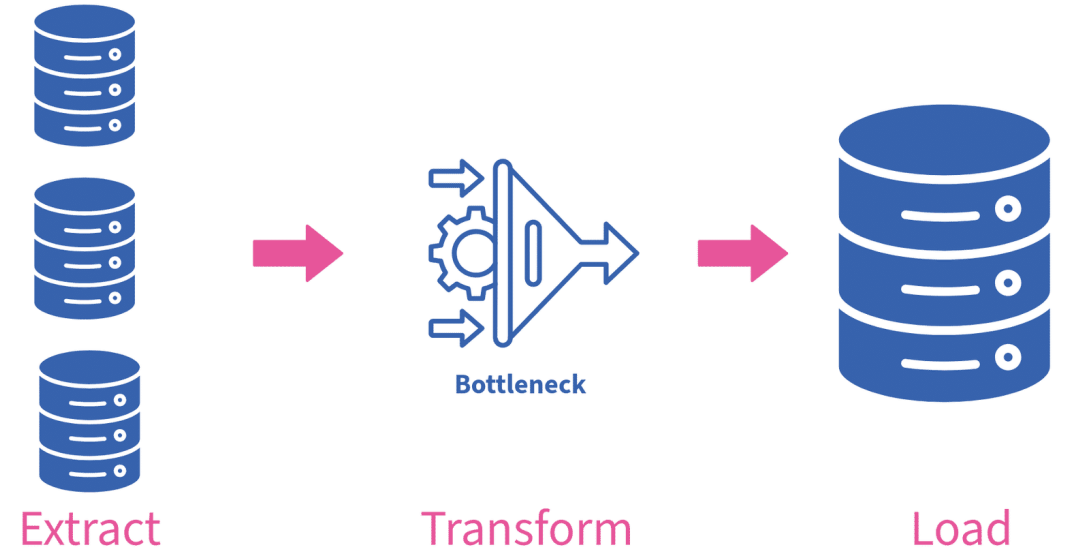
Recap of core concepts



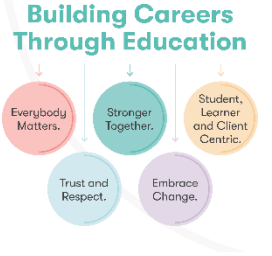
Why traditional ETL falls short

The limits of batch-based thinking

- ETL is slow, rigid, and batch-oriented
- Poor fit for real-time or streaming data
- Struggles with schema changes and unstructured formats
- Centralised transformations create bottlenecks



***From Rigid ETL to Modular Data Pipelines:
Embracing Flexibility and Orchestration***

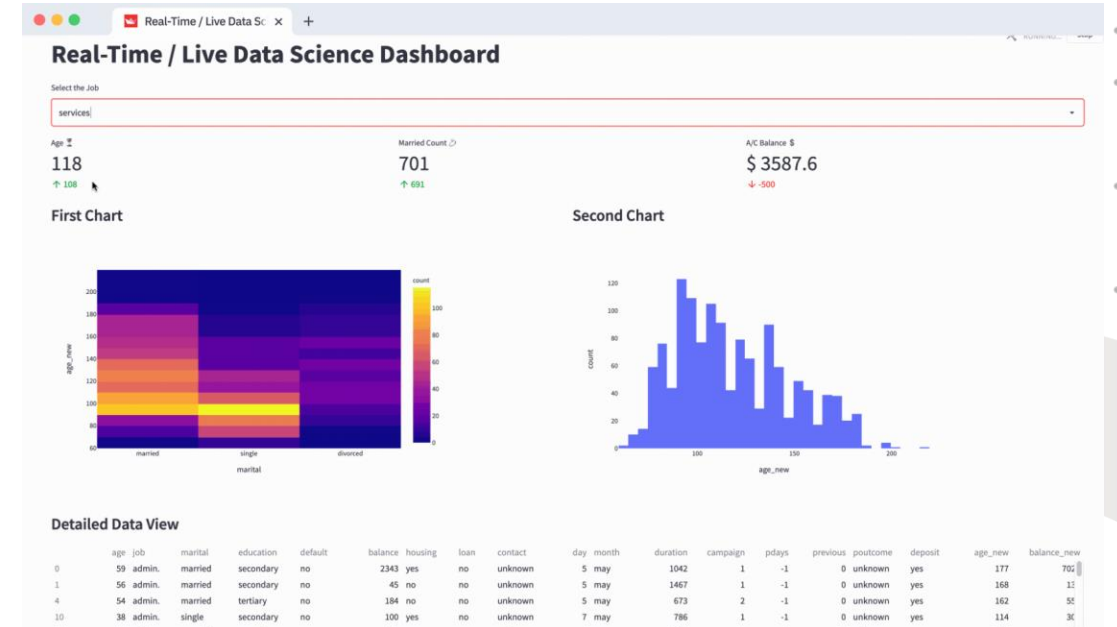
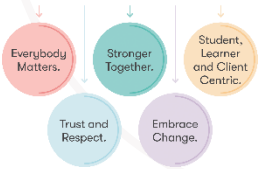


Modern integration methods

Flexible approaches for diverse needs

- **Federation:** Query data without moving it
- **Virtualisation:** Standardise views across sources
- **Blending:** Combine data quickly for specific use cases
- **ELT:** Load first, transform later for speed and scale

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Data federation example: A dashboard pulling live data from multiple APIs, image source: [Streamlit](#)

Real-time streaming pipelines

From snapshots to continuous flow

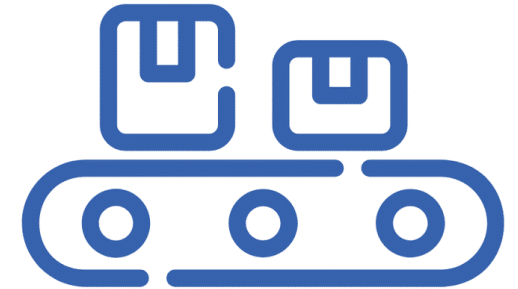
- Streaming enables low-latency, event-driven processing
- Ideal for dashboards, alerts, and automation
- Supports real-time sources like IoT, logs, and APIs
- Requires careful design: ordering, state, fault tolerance



Batch pipeline

Like a postal truck doing pickups and deliveries once a day.

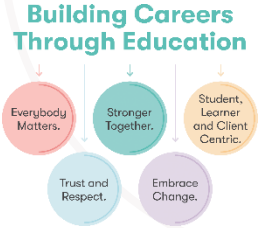
Vs



Streaming pipeline

Like a high-speed conveyor belt: always on, always moving.

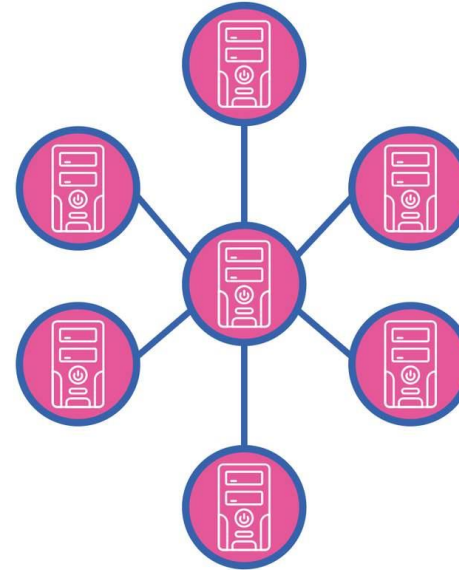
An analogy for the difference between batch and streaming pipelines



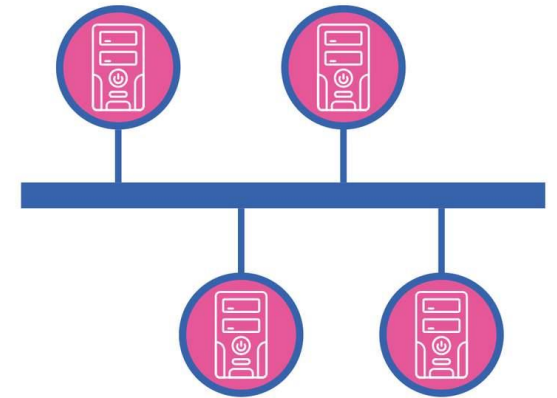
Integration architecture options

Designing for scale, resilience, and control

- **Point-to-Point:** Simple but hard to scale
- **Hub-and-Spoke:** Centralised control and governance
- **Distributed:** Modular, fault-tolerant, cloud-native
- Architecture impacts flexibility and maintainability



Hub and spoke



Bus

Figure: Hub-and-Spoke versus Bus Architecture



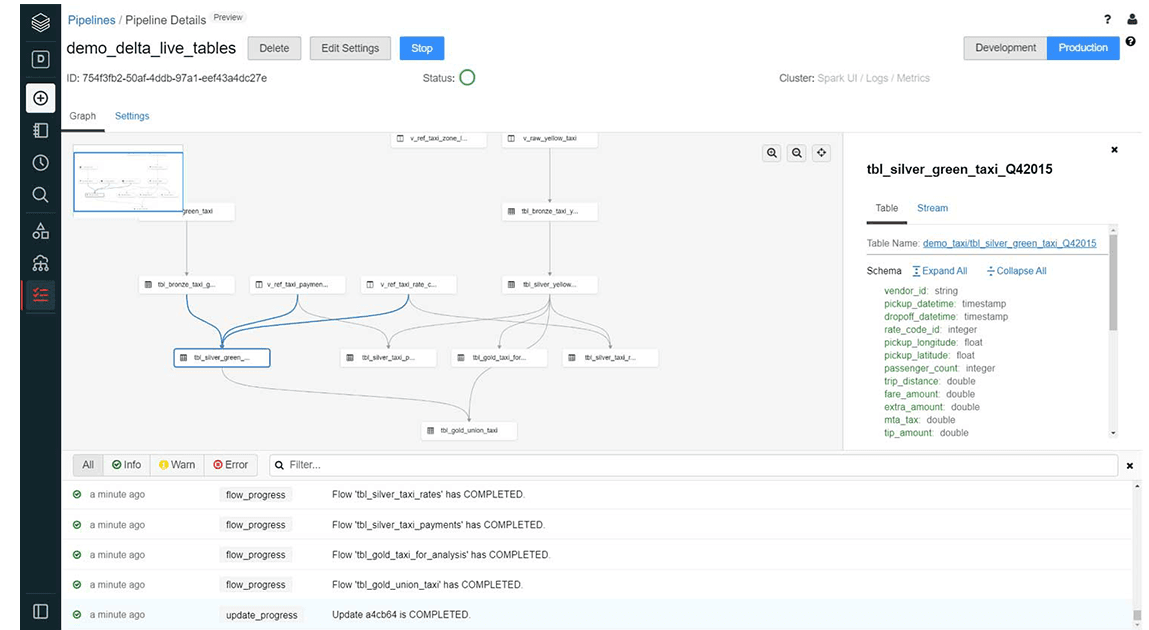
Consider your own organisation's needs - compliance, scale, agility - how these influence architectural decisions?



Tooling across the pipeline

The right tools for the right stage

- **NiFi:** Ingestion and routing
- **Kafka:** High-throughput streaming backbone
- **dbt/Talend:** Transformation and data quality
- **Airflow/Prefect:** Orchestration and scheduling

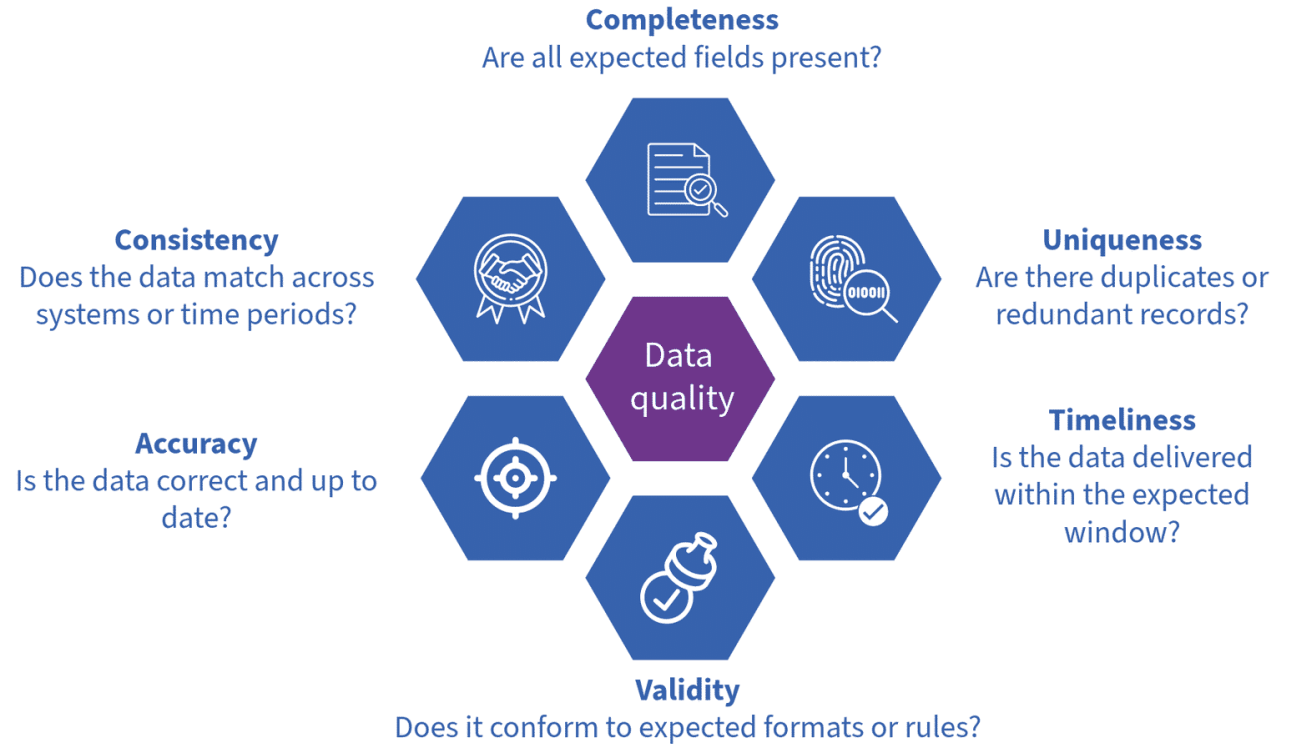


Modern Data Pipeline: Right Tools at Every Stage

Data quality is non-negotiable

Trustworthy data, reliable decisions

- Poor data = poor decisions and wasted effort
- Key dimensions: accuracy, completeness, consistency
- Pipelines must validate and monitor data continuously
- Quality issues can scale quickly if unchecked



Key Attributes of Data Quality in the Pipeline

Built-in governance

Visibility, Control, and Compliance

- Track data lineage and transformations
- Enforce schema validation and access control
- Log data movement and changes
- Governance is embedded, not optional

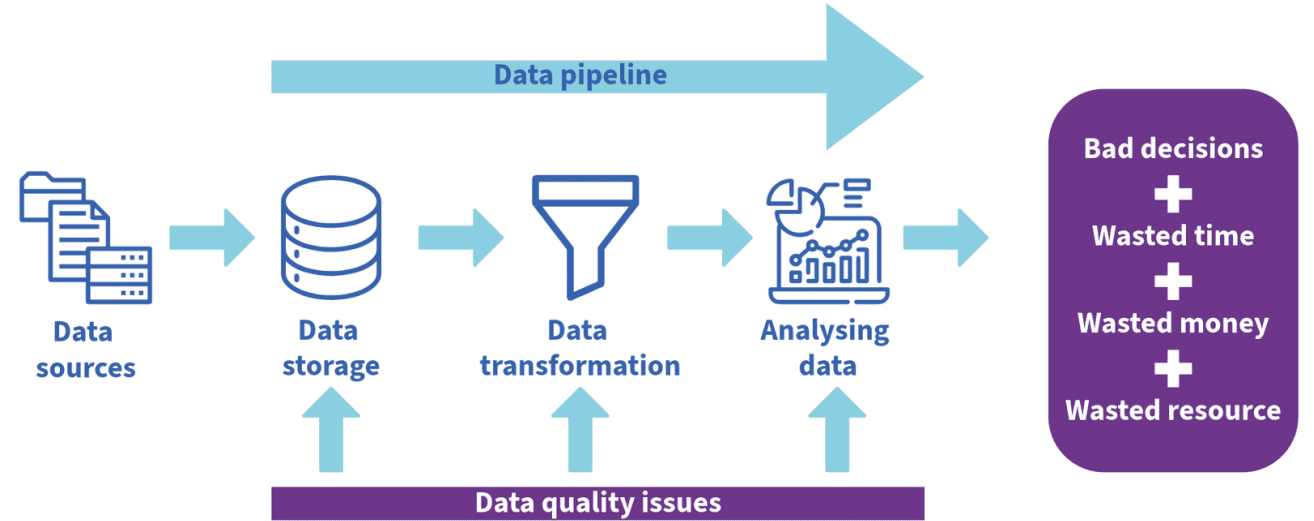
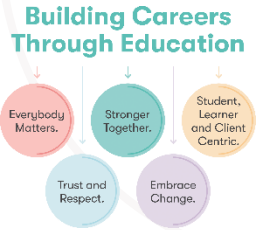


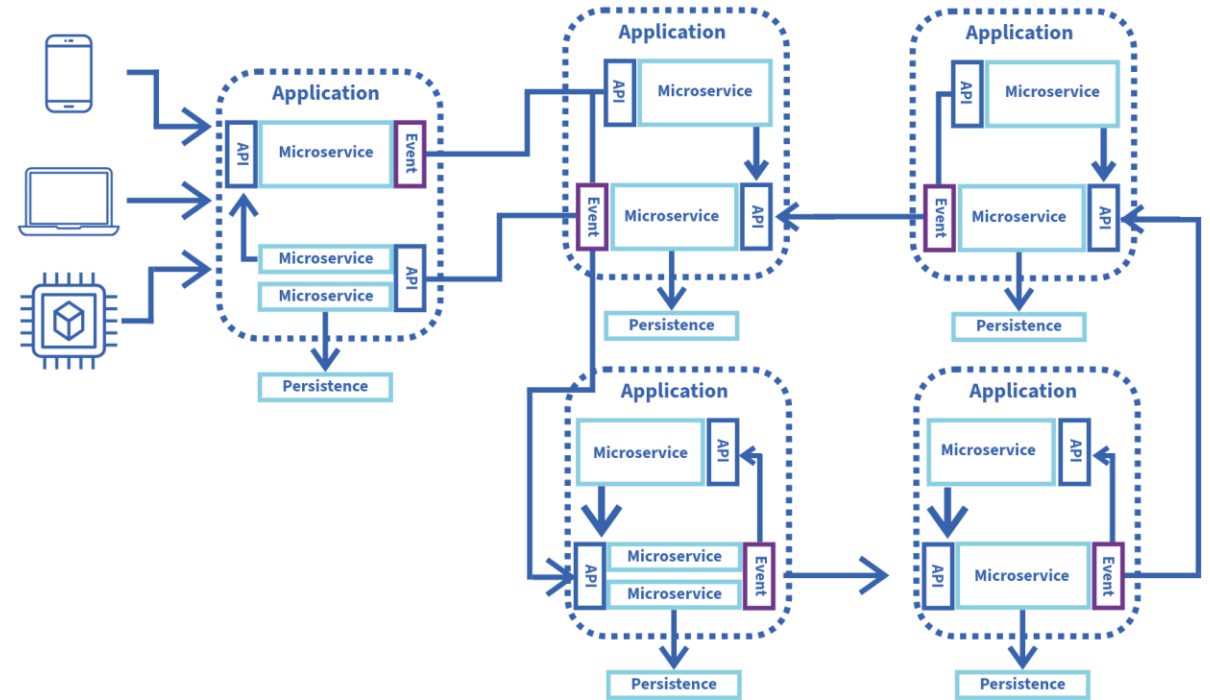
Figure: Data quality issues in pipelines



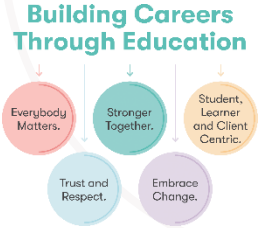
Hybrid, modular design thinking

No one-size-fits-all pipeline

- Track data lineage and transformations
- Enforce schema validation and access control
- Log data movement and changes
- Governance is embedded, not optional

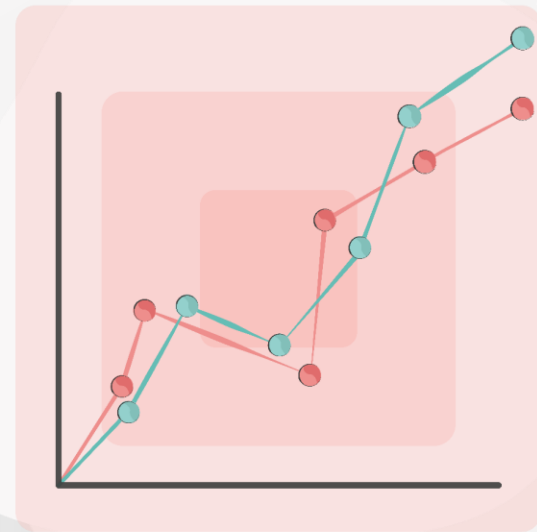


Hybrid, Modular Design for Scalable Data Governance





Practical lab



Exercise part 1

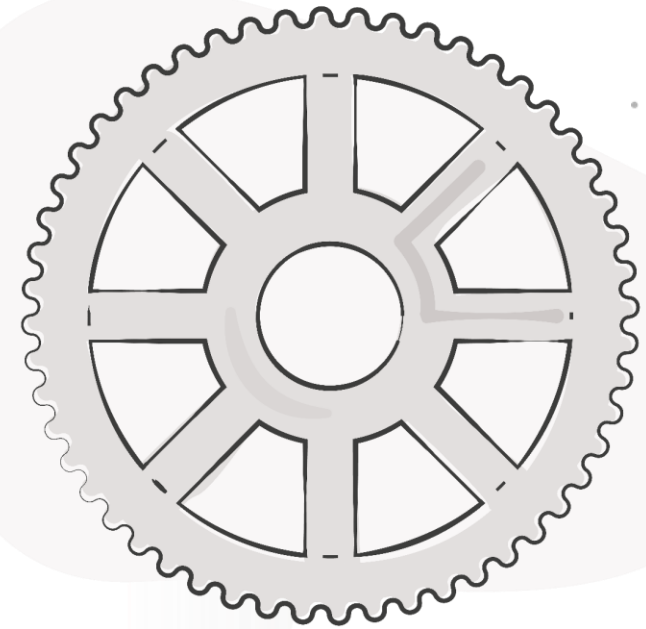
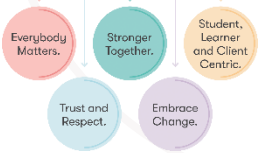
Initial Design Based on UK Data

1. **Design Database Schema:** Create a schema to receive user data from an existing software application.
2. **Implement Data Pipeline:** Develop a pipeline to clean, validate, and load data into the new schema.
3. **Use Test Data:** Utilise provided test data for 10 UK users and their login timestamps.
4. **Instructor Support:** Your instructor will act as the stakeholder SME for any data-related questions.
5. **Future Phases:** Prepare to integrate data from additional countries with varying formats.
6. **Documentation:** Establish comprehensive documentation for the schema and pipeline for future implementation by other teams.

Files Provided in the Hub:

- UK User Data.csv: Contains 10 sample records.
- UK-User-LoginTS.csv: Contains login timestamps for January 2025.

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Practical challenge

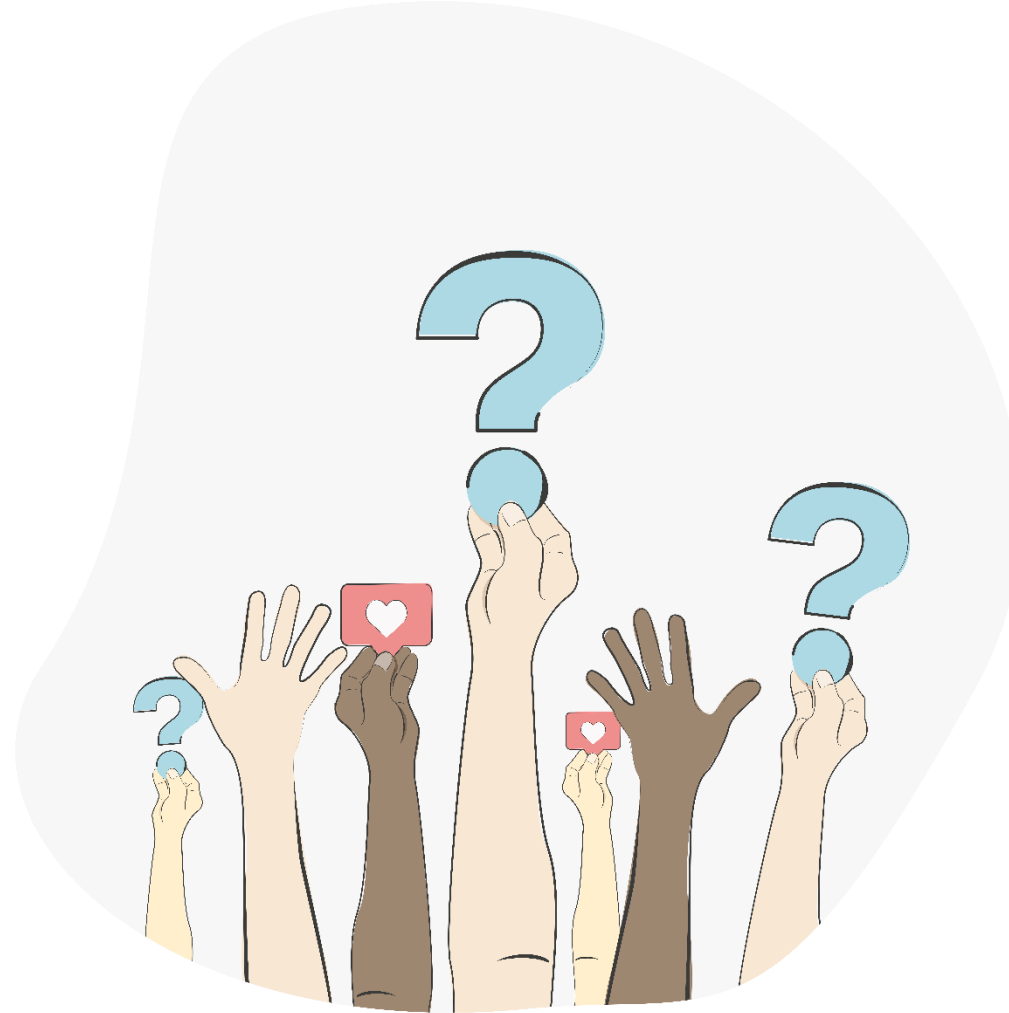
Key Learning Summary

Here are some key points to takeaway from this session:

- **Traditional ETL is limited** by latency, rigidity, and poor support for real-time, unstructured, or large-scale data needs.
- **Modern integration paradigms** include data federation, virtualisation, blending, and ELT - each suited to different pipeline goals and data types.
- **Streaming data integration** enables real-time insights using tools like Kafka and CDC, supporting event-driven architectures.
- **Architectural models** such as point-to-point, hub-and-spoke, and distributed integration impact scalability, fault tolerance, and governance.
- **Tooling ecosystems** (e.g., Apache NiFi, Talend, Kafka, dbt, Airflow) support ingestion, transformation, orchestration, and monitoring across the pipeline lifecycle.
- **Data quality and governance** are essential for reliable pipelines, using mechanisms like schema validation, lineage tracking, and automated quality checks.
- **Hybrid integration strategies** are often necessary - combining multiple techniques to balance speed, flexibility, and control.



Any questions or feedback?



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Thank you

